



**Department of Information Technology
Academic Year 2024– 2025 (Odd Semester)**

Degree, Semester & Branch : III Semester-B. Tech Information Technology

Course Code & Title : CS3391 Object Oriented Programming

Name of the Faculty member (s) : S. Sakkaravarthi, AP/IT

Innovative Practice Description

• **Unit / Topic : Unit 4 / Basic String class and Methods**

• **Course Outcome : CO4**

• **Topic Learning Outcome: TLO14**

• **Activity Chosen : Reflection**

• **Justification:**

- Reflection is a method that allows teachers to evaluate their teaching strategies and identify areas for enhancing student comprehension. During this activity, students are encouraged to write down any doubts or unclear concepts, enabling the teacher to review, address, and clarify those points effectively.
- This activity was specifically chosen to assess how well the students understood the Basic String Class and its methods in Java.

• **Time Allotted for the Activity: 15 Minutes**

• **Details of the Implementation:**

- A 15-minute reflection activity was conducted after teaching the features of String Class and its methods such as string length, string concatenation, and various string manipulation techniques using example programs in the classroom as shown in Fig 1.
- Students were asked to write down any questions or doubts related to string creation and manipulation on a sheet of paper.
- Students noted down their doubts related to the ways of string creation and the manipulation methods discussed in class, as shown in Fig. 2.
- The teacher reviewed the written doubts, prepared detailed responses, and distributed the answered sheets in the following class.
- This activity enabled students to overcome their difficulties and develop a clear understanding of the topic.

• **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO10	PSO1
CO4	3	2	3	2	3

(1 – Low 2 – Moderate 3 – High)

• PO / PSO mapped:

Innovative practice	PO1	PO2	PO3	PO10	PSO1
CO4	3	2	3	2	3
Justification for correlation	Students were introduced to the Basic String Class and its methods in Java, equipping them with the fundamental skills required for use in coding tasks and software projects.	Students will be able to apply string operations in real-time scenarios like form validation and data parsing.	Students will be able to write efficient programs that manipulate string data to solve practical problems.	Students will be able to improve their communication skills by expressing their doubts and reflections on string manipulation in Java.	Students can integrate string methods into larger Java applications, ensuring better software quality and performance.

• Images / Screenshot of the practice:



Fig 1: Reflection Activity

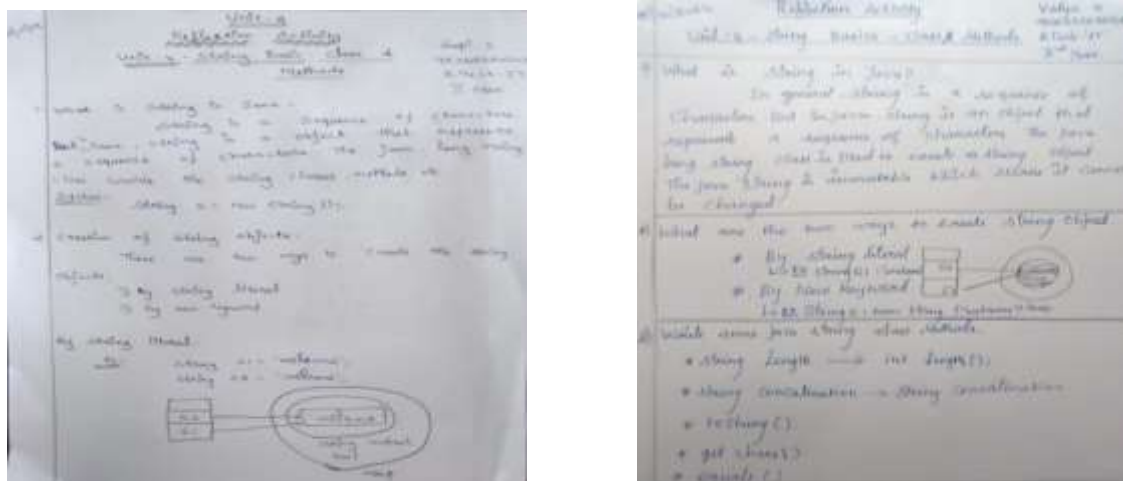


Fig 2: Sample Sheet of Reflection Activity

- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

- Students shared that the activity encouraged them to reflect on and assess their grasp of the concept.
- The activity created a comfortable environment for students to express and clarify their doubts related to string creation, manipulation, and built-in methods.
- Students improved their comprehension of how specific String methods work and how they can be used to perform efficient string processing.

- ❖ ***Benefit of the practice:***

- The reflection activity encouraged them to think about the logic behind using each String method, rather than just memorizing syntax.
- This activity made the slow learners to clarify the doubts easily.
- From this activity, students learned how specific String methods contribute to solving real-world problems involving text manipulation.

- ❖ ***Challenges faced in implementation:***

- Few Students were not able to express their doubts in the questions.
- Some students did not write any doubts, limiting the scope of feedback.

During the next class, I clarified the students' doubts, and even those who did not participate in the reflection activity mentioned that the discussion enhanced their grasp of the topic.

References:

- https://www.ritrjpm.ac.in/images/computer-science/2022-2023/IG_Reflection.pdf
- https://www.ritrjpm.ac.in/images/computer-science/2022-2023/1.KVS_Reflection_FDS.pdf

Signature of Faculty Member

HOD